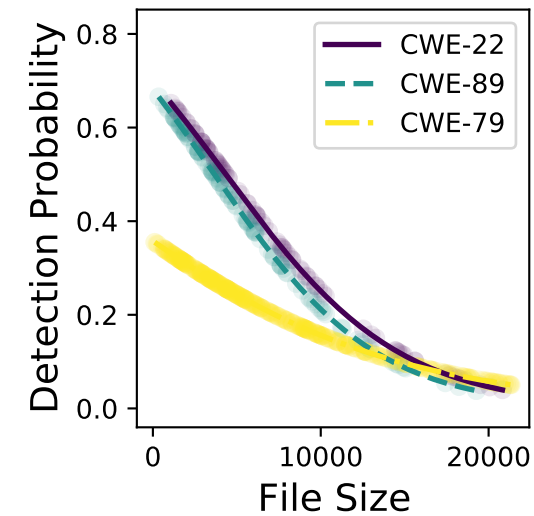
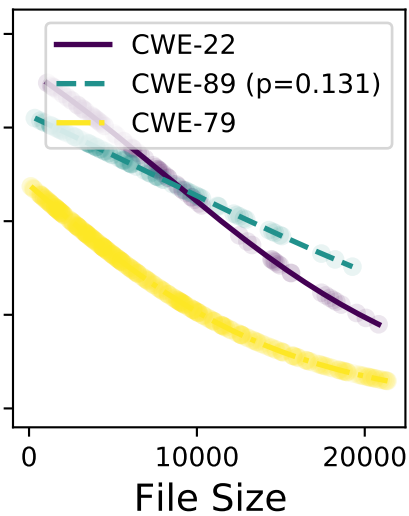


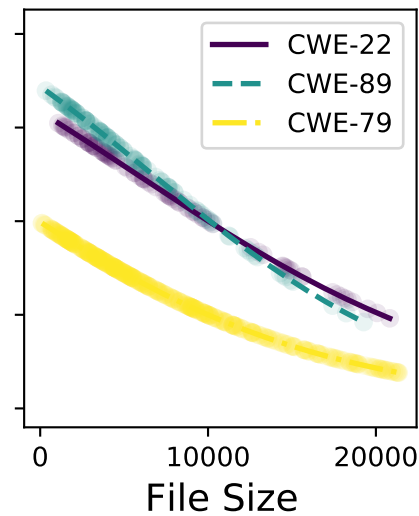
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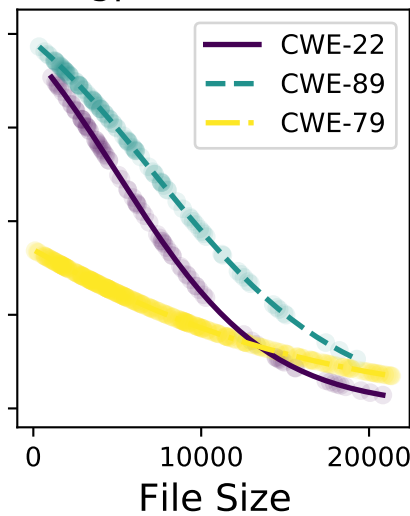
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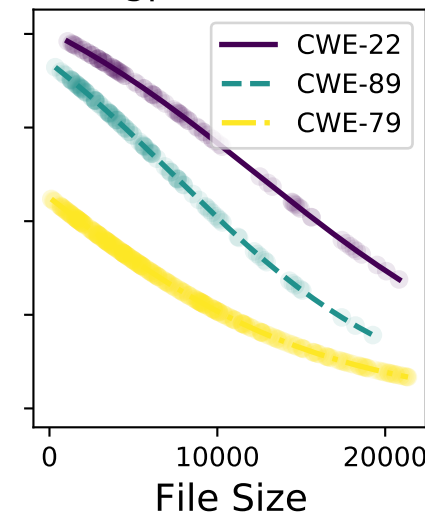
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gpt-3.5-turbo



gpt-4-turbo



gpt-4o

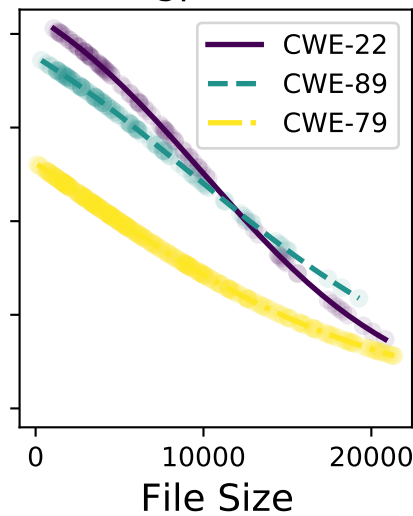


Figure 10 is a line graph titled "Detection Probability vs. Bug Position". The y-axis is labeled "Detection Probability" and ranges from 0.0 to 0.8. The x-axis is labeled "Bug Position" and ranges from 0 to 20,000. There are three data series: CWE-22 (solid purple line), CWE-89 (dashed teal line), and CWE-79 (dash-dot yellow line). CWE-22 and CWE-89 start at a detection probability of approximately 0.6 at bug position 0 and decrease sharply, reaching near 0.0 by bug position 10,000. CWE-79 starts at a lower detection probability of approximately 0.3 at bug position 0 and decreases more gradually, reaching near 0.0 by bug position 20,000. Shaded regions around the lines indicate confidence intervals.

The graph displays the distribution of bug positions for three CWE categories. The x-axis represents the 'Bug Position' from 0 to 20,000. The y-axis represents the frequency or count of bugs at each position. CWE-22 (purple solid line) shows a high frequency at position 0, which decreases as the position increases. CWE-89 (teal dashed line) shows a relatively flat distribution across the range. CWE-79 (yellow dash-dot line) shows a lower frequency at position 0, which also decreases as the position increases.

Bug Position	CWE-22	CWE-89 (p=0.852)	CWE-79
0	High	Medium	Low
10000	Medium	Medium	Low
20000	Low	Medium	Low

Line graph showing the number of bugs found versus Bug Position for three CWE categories: CWE-22 ($p=0.055$), CWE-89, and CWE-79. The x-axis represents Bug Position from 0 to 20,000. The y-axis represents the number of bugs found. CWE-22 (solid purple line) starts highest, followed by CWE-89 (dashed teal line), and CWE-79 (dotted yellow line) starts lowest. All three lines show a decreasing trend as bug position increases.

The graph displays the distribution of bug positions for three CWE types. The x-axis represents the 'Bug Position' from 0 to 20,000. The y-axis represents the 'Bug Position' from 0 to 20,000. The legend indicates three series: CWE-22 (purple solid line), CWE-89 (teal dashed line), and CWE-79 (yellow dotted line). All three series show a decreasing trend, with CWE-22 starting at the highest position and CWE-79 starting at the lowest position.

Bug Position	CWE-22	CWE-89	CWE-79
0	~15,000	~18,000	~8,000
5,000	~10,000	~12,000	~5,000
10,000	~7,000	~8,000	~4,000
15,000	~5,000	~5,000	~3,000
20,000	~4,000	~4,000	~3,000

A line graph showing the distribution of bug positions for three CWE categories: CWE-22 (purple solid line), CWE-89 (teal dashed line), and CWE-79 (yellow dash-dot line). The x-axis represents 'Bug Position' from 0 to 20,000. The y-axis represents frequency, with tick marks at 0, 0.5, 1, 1.5, and 2. CWE-22 has the highest frequency, peaking at approximately 1.8 at position 0. CWE-89 peaks at approximately 1.5 at position 0. CWE-79 has the lowest frequency, peaking at approximately 0.8 at position 0. All three curves show a decreasing trend as the bug position increases.

Bug Position	CWE-22	CWE-89	CWE-79
0	1.8	1.5	0.8
5000	1.2	1.0	0.5
10000	0.8	0.6	0.3
15000	0.5	0.3	0.2
20000	0.3	0.2	0.1

Figure 10: Relationship between Bug Position and CWE. The graph plots the frequency of bugs (y-axis) against their position in the code (x-axis, 0 to 20,000). Three CWE categories are compared: CWE-22 (solid purple line), CWE-89 (dashed teal line, $p=0.337$), and CWE-79 (dash-dot yellow line). CWE-22 shows a steep decline in frequency as position increases. CWE-89 shows a more gradual decline. CWE-79 starts at a lower frequency and declines more gradually than CWE-22.